L1.4 The Derivative

Work in your group. Check your answers against the Quarto tonight.

1 · The derivative at a point

> **Stop.** Write the difference quotient from memory before you substitute anything into it.

1. $f(x) = 2x^2 + 1$. Find $f'(2)$ **two ways**, and check that they agree.

(a) Put 2 in at the start and take the limit.

(b) Find $f'(x)$ first, then put 2 in at the end.

2. Now give the **equation of the tangent line** to $f(x) = 2x^2 + 1$ at $x = 2$.

What extra piece of information did you need that part (a) above did not give you?

3. A particle's position is $x(t) = 5t^2$.

(a) Find its **average** velocity on the interval from $t = 5$ to $t = 5 + h$. Leave h in your answer.

(b) Now find its **instantaneous** velocity at $t = 5$.

(c) Which one of those is a secant slope and which is a tangent slope?

2 · The derivative as a function

4. Find $\dfrac{dy}{dx}$ for $y = \sqrt{x}$ **from the definition**, then evaluate $\left.\dfrac{dy}{dx}\right|_{x=3}$.

5. $f(x) = x^2 - 4$. Find $f'(x)$ and then $f''(x)$, both **from the definition**. Sketch all three stacked on the grids below.

f

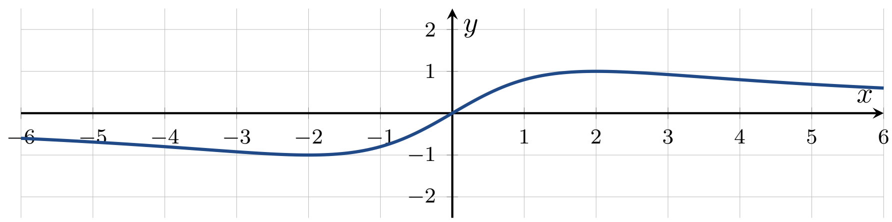
f'

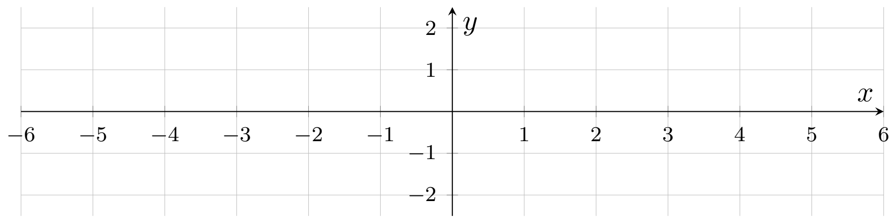
f''

Stop. What does a horizontal tangent on f become on f'? Write it down before you sketch.

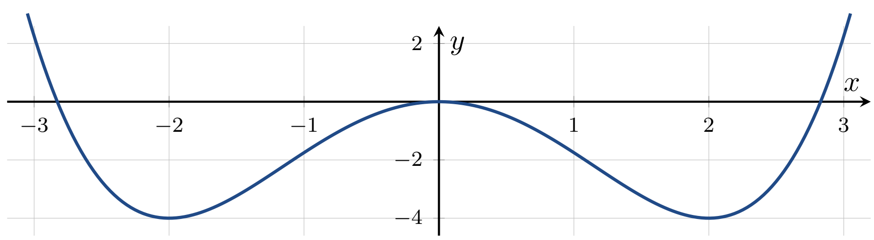
6. Study the graph of f below — where it is increasing, where it is decreasing, and where it has horizontal tangent lines. Then **sketch** f' on the empty grid.

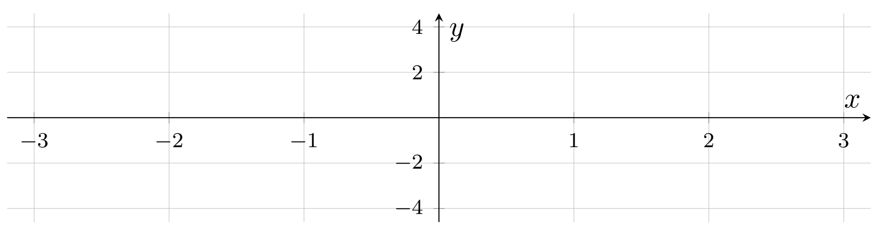
Where f has a horizontal tangent, $f' = 0$. Where f is increasing, $f' > 0$ — above the axis. Where f is decreasing, $f' < 0$ — below the axis. And find the inflection points.





7. Same task, harder. **Mark the inflection points first**, then the horizontal tangents, then sketch f'.

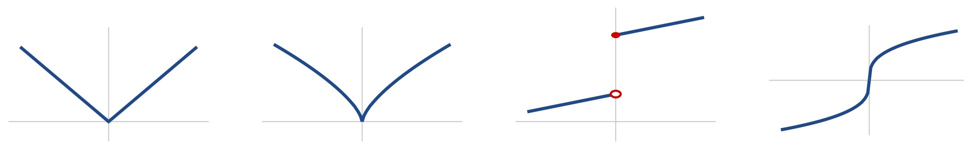




How many horizontal tangents does this one have, and how many inflection points?

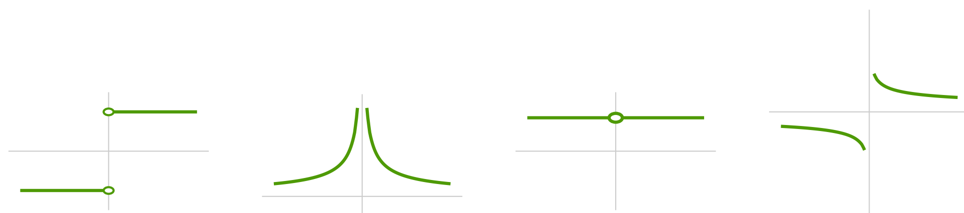
8. **Match** each graph of f to the graph of its f'. Then name the kind of failure in each.

The functions.



A **B** **C** **D**

The derivatives.



I **II** **III** **IV**

A $\to$ ______ B $\to$ ______ C $\to$ ______ D $\to$ ______

9. $f(x) = |x|$.

(a) Write f as a piecewise function, then write f' as one.

(b) Show f is **not differentiable** at 0.

(c) **Is f continuous at** 0**?** Answer it, and say what that tells you about the two words.

10. $f(x) = |x^2 - 4|$.

(a) Write f piecewise. *Solve* the inequality — do not guess where the branches switch.

(b) Write f' piecewise.

(c) State the set on which f is differentiable.

11. **BOARDS** Find $f'(x)$ for $f(x) = \dfrac{1}{x}$ from the definition. Then give the equation of the tangent line at $x = 2$.